

AK-KUDUK, Kazakhstan

WMO: 382320

Lat: 42.9713N

Lon: 54.0945E

Elev: 78

StdP: 100.39

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.2	-21.9	0.5	-16.0	-19.4	0.7	-13.2	10.4	-1.4	9.8	-0.9	1.9	20	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	40.6	19.5	39.0	19.2	37.5	18.8	22.1	34.3	21.1	33.5	20.2	33.1	4.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.3	26.8	16.7	12.0	26.4	15.4	11.1	25.9	65.0	33.8	61.4	33.4	58.3	33.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.6	8.5	-21.0	43.2	5.0	1.8	-24.6	44.4	-27.5	45.5	-30.3	46.5	-33.9	47.7		
			-21.1	23.8	4.8	1.7	-24.6	25.1	-27.4	26.0	-30.2	27.0	-33.7	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	-3.2	-1.1	6.0	13.9	21.4	27.2	30.0	28.3	20.8	12.4	4.4	-1.1
	DBStd	12.56	5.67	5.62	4.65	4.96	4.29	3.68	3.25	3.56	4.06	4.72	5.02	5.03
	HDD10.0	1427	408	311	139	18	0	0	0	0	0	30	177	344
	HDD18.3	2994	667	543	383	148	20	1	0	0	21	191	419	603
	CDD10.0	2644	0	1	15	137	354	517	619	568	323	103	8	0
	CDD18.3	1170	0	0	1	16	116	268	361	310	94	6	0	0
	CDH23.3	16026	0	0	6	199	1314	3661	5312	4354	1092	87	0	0
	CDH26.7	9136	0	0	1	57	572	2107	3301	2622	463	13	0	0
Wind	WSAvg	3.5	3.1	3.5	4.0	4.0	3.8	4.0	3.6	3.2	3.3	3.2	3.0	3.2
Precipitation	PrecAvg	144	10	8	20	24	18	11	8	3	10	9	12	12
	PrecMax	321	24	44	72	137	62	50	63	14	51	29	31	33
	PrecMin	62	0	0	1	0	0	0	0	0	0	0	0	0
	PrecStd	57	6	9	17	29	17	12	13	4	11	10	9	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	14.1	23.2	31.5	36.5	41.1	42.9	42.4	37.6	28.5	19.6	11.8
		MCWB	6.3	7.6	11.9	15.7	17.5	19.2	19.7	19.7	18.4	14.8	11.9	6.9
	2%	DB	6.6	10.5	19.2	28.2	34.1	39.0	40.9	40.2	33.6	25.8	16.6	8.5
		MCWB	4.5	6.3	10.2	14.8	16.7	18.6	19.8	19.3	16.8	14.2	10.5	5.8
	5%	DB	4.8	7.8	16.2	25.6	31.9	37.4	39.3	38.5	31.4	23.4	13.9	6.7
		MCWB	3.2	4.6	8.7	13.8	16.3	18.2	19.5	19.0	16.2	13.1	8.9	4.6
	10%	DB	3.2	5.8	13.7	22.9	30.1	35.7	37.7	36.8	29.2	21.0	11.5	5.0
		MCWB	1.9	3.2	7.6	12.5	16.0	18.0	19.2	18.5	15.4	12.0	7.8	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	8.5	12.6	17.1	20.1	22.4	23.7	23.9	20.5	17.6	13.5	7.9
		MCDB	9.3	13.0	20.8	27.1	30.0	33.7	35.3	33.8	31.3	23.8	17.5	10.1
	2%	WB	5.0	6.7	11.0	15.8	18.6	21.0	22.3	21.7	18.6	15.7	11.4	6.3
		MCDB	6.4	10.0	17.7	25.2	28.4	33.3	34.6	34.3	29.0	22.3	15.3	8.2
	5%	WB	3.3	5.0	9.5	14.7	17.7	20.0	21.4	20.5	17.4	14.3	9.6	4.9
		MCDB	4.5	7.4	15.5	23.2	27.9	32.6	33.9	33.6	28.3	21.3	12.8	6.4
	10%	WB	2.1	3.5	8.1	13.5	16.9	19.1	20.6	19.5	16.4	13.0	8.1	3.6
		MCDB	3.2	5.5	12.8	21.6	27.3	31.8	33.4	33.3	26.9	19.6	11.1	4.8
Mean Daily Temperature Range		MDBR	7.4	8.9	10.9	12.1	12.4	13.0	12.6	13.6	13.8	12.9	9.3	7.4
		MCDBR	7.9	11.7	15.4	15.7	14.7	15.3	14.5	16.1	16.5	16.5	13.3	9.6
		MCWBR	6.1	7.7	8.0	6.7	5.4	5.4	5.0	6.1	7.3	8.5	8.2	7.2
	5% WB	MCDBR	7.5	10.7	14.1	14.1	12.9	13.0	12.4	13.6	14.1	13.9	11.4	8.4
		MCWBR	6.0	7.3	7.6	6.4	5.3	5.6	5.4	6.1	7.1	8.1	7.5	6.7
Clear-Sky Solar Irradiance	taub	0.388	0.413	0.465	0.536	0.523	0.505	0.507	0.484	0.455	0.439	0.401	0.391	
	taud	2.041	1.995	1.908	1.773	1.851	1.919	1.926	1.968	1.995	2.006	2.067	2.044	
	Ebn at Noon	693	750	760	739	765	780	773	776	762	710	672	644	
	Edh at Noon	112	138	171	211	201	188	185	172	155	135	109	103	
All-Sky Solar Radiation	RadAvg	1.40	2.45	3.77	5.04	6.26	7.12	6.75	6.20	4.93	3.23	1.95	1.26	
	RadStd	0.22	0.31	0.34	0.49	0.41	0.30	0.29	0.30	0.29	0.26	0.23	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+1.00	-1.07	-0.98	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page